DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC ASSIGNMENT

CO4 - AT 3

COMPARATIVE ANALYSIS

Network Topology Design for Real-World Scenarios

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Declaration

I hereby declare that this assignment is my own original work, prepared under the guidance of my course faculty, and has been submitted in accordance with the academic requirements of the course.

Question 1: Compare Star and Bus Topologies in Cost, Reliability and Performance

Star topology and bus topology are both used to connect devices in a network, but they differ significantly in cost, reliability and performance. In a star topology, every device has an individual connection to a central switch or hub. In a bus topology, all devices share a single backbone cable. The choice between them depends on the size of the network, budget and required level of reliability.

Cost Comparison

Star topology is generally more expensive because it requires a central networking device and a separate cable from every node to that device. As the number of systems increases, the amount of cabling and the number of switch ports also increase. Bus topology is cheaper to install because it mainly uses one common backbone cable and needs fewer individual connections. Therefore, bus topology is suitable when minimum installation cost is the main requirement.

Reliability Comparison

Star topology provides better reliability. If the cable or network interface of one node fails, only that particular node is affected and the remaining devices continue to communicate. However, the central switch or hub is an important dependency; if it fails, the connected network can stop. In bus topology, the backbone cable is the critical component. A failure or break in the main cable can disrupt communication for the entire network, making fault isolation more difficult.

Performance Comparison

Star topology usually provides better performance because communication is controlled through the central device and each node has a dedicated connection. This reduces the effect of collisions and makes network traffic easier to manage. In bus topology, all devices use the same shared medium. As more devices are added and traffic increases, collisions and waiting time can increase, resulting in reduced performance.

In conclusion, bus topology has the advantage of lower initial cost and simple installation, while star topology offers better reliability, easier troubleshooting and better performance. For most networks where stable communication and future expansion are important, star topology is the stronger choice.

Question 2: Compare Mesh and Star Topologies for Fault Tolerance

Fault tolerance is the ability of a network to continue operating even when a link or device fails. Mesh topology and star topology provide different levels of fault tolerance because their connection structures are different. Mesh topology is designed with multiple communication paths, whereas star topology depends on one central device for communication between nodes.

Mesh Topology

In a mesh topology, each node is connected to several other nodes, and in a full mesh every node can have a direct connection to every other node. These multiple paths provide very high fault tolerance. If one cable or connection fails, data can be transmitted through another available path. As a result, a single link failure normally does not stop communication between devices. This makes mesh topology highly reliable and suitable for critical environments where continuous availability is important.

Star Topology

In a star topology, every node is connected independently to a central switch or hub. It provides good fault isolation at the individual node level. If the cable connected to one computer fails, the other computers remain active because their connections are separate. However, the central device acts as a single point of failure. If the switch or hub stops functioning, communication between all connected nodes can be interrupted.

Comparison and Analysis

Mesh topology is more robust because communication does not depend on only one path. It can tolerate multiple link problems better than star topology. Star topology provides moderate fault tolerance because individual link failures are isolated, but the failure of the central device has a much larger effect. The advantage of star topology is that its structure is simpler and easier to manage, while mesh topology requires more links and has greater complexity.

Therefore, mesh topology is the better option when maximum reliability and fault tolerance are the main priorities. Star topology is suitable when a balance of reliability, simplicity and manageable infrastructure is required. In critical networks, the multiple alternate paths of mesh topology provide a clear reliability advantage over star topology.

Question 3: Compare Bus and Mesh Topologies Based on Installation Complexity

Installation complexity refers to the amount of cabling, networking connections, time and effort required to build a network. Bus topology and mesh topology are opposite in this aspect. Bus topology has a very simple structure, whereas mesh topology becomes increasingly complex as the number of devices grows.

Bus Topology Installation

Bus topology uses a single backbone cable that is shared by all connected devices. Because only one main communication line is required, the number of cable connections is limited and the physical layout is straightforward. This makes bus topology relatively easy and quick to install for a small network. The lower number of connections also reduces the initial hardware requirement and installation cost.

Mesh Topology Installation

Mesh topology requires many direct connections between nodes. In a full mesh, every device must be connected to every other device. Therefore, each newly added device can require several additional cables and ports. As the network grows, the number of physical links increases rapidly, and cable management becomes more difficult. Installation also requires additional planning to ensure that all required connections are correctly established.

Complexity and Cost

Because of its simple shared backbone, bus topology needs less cabling and fewer connection points. It is therefore easier to implement when the network contains only a small number of devices. Mesh topology requires a much larger quantity of cabling and network interfaces. This makes installation more time-consuming and costly. The complexity also increases future maintenance because technicians must identify and manage a large number of individual links.

In conclusion, bus topology is much easier to install and is suitable for small networks where simplicity and low cost are important. Mesh topology is difficult and expensive to install, but this additional complexity provides multiple communication paths and high reliability. Therefore, the selection depends on whether the priority is simple installation or strong fault tolerance.

Question 4: Compare Hybrid and Star Topologies Based on Scalability

Scalability is the ability of a network to expand when the number of users, devices or network segments increases. Both hybrid and star topologies can be expanded, but hybrid topology provides greater flexibility because it combines two or more network topologies and can be adapted to different organizational requirements.

Scalability of Hybrid Topology

A hybrid topology may combine star, mesh, bus or other topology structures within the same organization. This allows a network administrator to select an appropriate design for each section of the network. When the organization expands, a new segment can be added without completely redesigning the existing network. For example, an additional star segment can be connected to the existing hybrid structure while the current segments continue to operate. This modular nature gives hybrid topology strong scalability.

Scalability of Star Topology

Star topology is also easy to expand because a new computer can normally be connected by adding another cable to the central switch or hub. However, its scalability is directly limited by the capacity of the central device. When all available ports are occupied, additional hardware such as another switch is required. Therefore, expansion is simple within the available port capacity but becomes more dependent on additional central devices as the network grows.

Comparison

Hybrid topology provides greater scalability because different network sections can be expanded independently and different topology types can be selected according to changing needs. Star topology is highly convenient for a single department or smaller network, but large expansion depends on the number and capacity of central switches. Hybrid topology can support larger and more complex organizational structures because it is not restricted to one topology design throughout the network.

Hence, star topology provides good scalability for simple networks, while hybrid topology is more suitable for large organizations that expect continuous growth. Its ability to add new network segments while preserving the existing structure makes hybrid topology the more flexible and scalable solution.

Question 5: Compare Maintenance Requirements of Star, Mesh, Bus and Hybrid Topologies

Network maintenance includes identifying faults, replacing failed links, managing connections and keeping communication stable. The maintenance effort differs considerably among star, mesh, bus and hybrid topologies because each topology has a different physical and logical structure.

Star Topology

Star topology is generally the easiest to maintain. Every device has a separate connection to the central switch or hub, so a problem can usually be isolated to a particular node or cable. If one cable fails, other devices are normally unaffected. The centralized structure also makes troubleshooting and network management more straightforward. The main point that requires attention is the central device because its failure can affect the whole star network.

Mesh Topology

Mesh topology offers excellent reliability, but it is more difficult to maintain. It contains a large number of connections between nodes, especially in a full mesh. When the network becomes large, technicians must monitor and troubleshoot many individual links. Although alternate paths keep communication active during failures, the large number of cables and ports increases maintenance complexity.

Bus and Hybrid Topologies

Bus topology has fewer components, but troubleshooting can be difficult because all devices depend on the common backbone. A fault in the main cable can affect the entire network, and locating the exact failure point may take time. Hybrid topology has moderate to high maintenance complexity because it combines several topology types. Each section may have different troubleshooting procedures, so careful monitoring and skilled administration are required.

Overall Comparison

Among the four designs, star topology is generally the easiest to maintain because faults can be isolated quickly. Bus topology is physically simple but backbone faults are difficult to locate. Mesh topology is highly reliable but requires more maintenance because of its many connections. Hybrid topology provides flexibility but needs administrators who understand all the combined structures. Therefore, maintenance requirements must be considered along with cost, reliability and scalability when selecting a topology.